

Strategies for preventing type 2 diabetes: an update for clinicians

Diabetes is a major and growing public health challenge which threatens to overwhelm medical services in the future. Type 2 diabetes confers significant morbidity and mortality, most notably with target organ damage to the eyes, kidneys, nerves and heart. The magnitude of cardiovascular risk associated with diabetes is best illustrated by its position as a coronary heart disease risk equivalent. Complications related to neuropathy are also vast, often working in concert with vascular abnormalities and resulting in serious clinical consequences such as foot ulceration. Increased understanding of the natural history of this disorder has generated the potential to intervene and halt pathological progression before overt disease ensues, after which point management becomes increasingly challenging. The concept of prediabetes as a formal diagnosis has begun to be translated from the research setting to clinical practice, but with continually updated guidelines, varied nomenclature, emerging pharmacotherapies and an ever-changing evidence base, clinicians may be left uncertain of best practice in identifying and managing patients at the prediabetic stage. This review aims to summarize the epidemiological data, new concepts in disease pathogenesis and guideline recommendations in addition to lifestyle, pharmacological and surgical therapies targeted at stopping progression of prediabetes to diabetes. While antidiabetic medications, with newer anti-obesity medications and interventional bariatric procedures have shown some promising benefits, diet and therapeutic lifestyle change remains the mainstay of management to improve the metabolic profile of individuals with glucose dysregulation. New risk stratification tools to identify at-risk individuals, coupled with unselected population level intervention hold promise in future practice.